Boson–Fermion Correspondence and the Jacobi Identity

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These notes are based on Chapters 4–5 of *Solitons: Differential Equations, Symmetries and Infinite Dimensional Algebras.*

1 Bosonic and fermionic algebras and the Fock space

We discuss free massless bosons and fermions on a $1+1$-dimensional cylinder. (If you do not understand this sentence, you may ignore it.)

1.1 Bosons

Let $x_1, \dots, x_n, \dots$ be formal variables. The Fock space is defined as $\mathbb{C}[x_1, \dots, x_n, \dots]$. Define a grading on the Fock space by $\mathrm{gr}(x_n) = n$ (physically this grading equals the energy).

The bosonic algebra consists of operators $a_n, a_n^\dagger$ satisfying $[a_n, a_m] = [a_n^\dagger, a_m^\dagger] = 0$, $[a_n, a_m^\dagger] = \delta_{m,n}$.

The action of the bosonic algebra on the Fock space is $a_n f = x_n f$, $a_n^\dagger f = \partial f / \partial x_n$.

A bit of physics: in the Fock space, 1 corresponds to the vacuum state, and x_n corresponds to an n-mode excitation.

1.2 Fermions

The fermionic case is a bit more involved. First write down the fermionic algebra: it is generated by $\psi_n, \psi_n^\dagger$ ($n \in \mathbb{Z} + \frac{1}{2}$) subject to the anticommutation relation $\{\psi_n, \psi_m^\dagger\} = \delta_{m+n,0}$, with all other anticommutators equal to 0.

Next define the Fock space: imagine placing black and white stones at the positions $\mathbb{Z} + \frac{1}{2}$ on a line. A basis consists of those configurations with black stones at all positions $n \gg 0$ and white stones at all positions $n \ll 0$. The vacuum state has black stones at all positions $n > 0$ and white stones at all positions $n < 0$. The condition above amounts to differing from the vacuum state by finitely many stones.

The action of the fermionic algebra on the Fock space: ψ_n turns the white stone at $-n$ into a black stone (and gives 0 if it is already black), while $\psi_n^\dagger$ turns the black stone at n into a white stone (and gives 0 if it is already white). This space is denoted by $\mathcal{F}$.

A note on the sign convention: each basis vector is obtained by applying some of the $\psi_n, \psi_n^\dagger$ in order to the vacuum state. If one acts on it with another operator, one must commute that operator past the others to obtain the correct ordering. See the reference for details.

Introduce two gradings on the Fock space: the particle number $\text{charge}(\psi_n) = 1$, $\text{charge}(\psi_n^\dagger) = -1$, and the energy $\text{energy}(\psi_n) = -n$, $\text{energy}(\psi_n^\dagger) = -n$. The corresponding graded piece is denoted by $\mathcal{F}_d^{(l)}$.

2 The boson–fermion correspondence

Theorem 1. *There is an isomorphism from the bosonic Fock space to $\mathcal{F}^{(l)}$ that maps a state of energy n to a state of energy $n + l^2/2$.*

We can shift a state of particle number l to the right by l, reducing the problem to the case $l = 0$. The book only gives the map from $\mathcal{F}^{(0)}$ to the bosonic Fock space, and proves that it is surjective. The isomorphism is proved via the Jacobi identity. (I have always felt that there should be an approach avoiding the Jacobi identity or Schur polynomials.)

2.1 Φ and Φ^{-1}

First define $H_n := \sum_{j \in \mathbb{Z}+1/2} :\psi_{-j}\psi_{j+n}^\dagger:$. Since in these notes $n \neq 0$ in general, the normal ordering has no effect.

Define $\Phi : \mathcal{F}^{(0)} \to \mathbb{C}[x_1, \dots, x_n, \dots]$ by

$$|u\rangle \mapsto \langle 0| \exp\Big(\sum_{i \geq 1} H_i x_i\Big) |u\rangle .$$

We can compute some examples: (TBD)

Theorem 2.

$$\Phi(H_n |u\rangle) = \begin{cases} \frac{\partial}{\partial x_n}\Phi(|u\rangle) & n > 0, \\ -n x_n \Phi(|u\rangle) & n < 0. \end{cases}$$

The proof is by the commutation relations.

By repeatedly using the relation for $n < 0$, we can prove that Φ is surjective. Formally we may write

$$\Phi^{-1}(F) = F(-H_1, -H_2/2, \dots) |0\rangle .$$

3 The Jacobi identity

Let $\mathcal{F}_d^{(l)}$ denote the subspace of the fermionic Fock space with particle number l and energy d, and define the generating function

$$\text{ch}(\mathcal{F}) = \sum_{l \in \mathbb{Z},\, d - l^2/2 \in \mathbb{Z}^{\geq 0}} \dim(\mathcal{F}_d^{(l)})\, z^l q^d.$$

On the fermionic side, since the fermionic Fock space is generated by creation operators,

$$\text{ch}(\mathcal{F}) = \prod_{j \in \mathbb{Z}^{\geq 0} + 1/2} (1 + zq^j)(1 + z^{-1}q^j).$$

On the bosonic side,

$$\dim(\mathcal{F}_d^{(l)}) = p(d - l^2/2),$$

where $p(n)$ is the partition function, satisfying $\sum_{n\geq 0} p(n)q^n = \prod_{m\geq 1}(1-q^m)^{-1}$. Making the change of variables $d \to d - l^2/2$ (there seems to be a typo in the book),

$$\mathrm{ch}(\mathcal{F}) = \sum_{l\in\mathbb{Z}} z^l q^{l^2/2} \prod_{m\geq 1}(1-q^m)^{-1}.$$

Rearranging further, and making the change of variables $z \to -zq^{1/2}$,

$$\prod_{m\geq 1}(1-q^m)(1-zq^{m-1})(1-z^{-1}q^m) = \sum_{m\in\mathbb{Z}}(-z)^m q^{m(m-1)/2}.$$